PLANETS Database using Prolog

Aim

To write and execute a Prolog program to create a **PLANETS database** and retrieve the properties of a particular planet using suitable queries.

Algorithm

1. Start the program.
2. Define facts for different planets using the planet/5 predicate.
3. Store the **planet name, type, size, temperature, and position from the Sun** as properties.
4. Define the planet_properties/5 predicate to retrieve the properties of a given planet.
5. Enter a query with the required planet name.
6. Prolog searches the database and matches the given planet with the stored facts.
7. Display the corresponding planet properties.
8. Queries can also be used to find planets based on their type, size, or temperature.
9. Stop the program.

Code

```
% Prolog Program for PLANETS DB

% Facts about planets and their properties

planet(mercury, rocky, small, hot, closest_to_sun).

planet(venus, rocky, small, hot, 2nd_closest_to_sun).

planet(earth, rocky, medium, temperate, 3rd_closest_to_sun).

planet(mars, rocky, small, cold, 4th_closest_to_sun).

planet(jupiter, gas_giant, large, cold, 5th_closest_to_sun).

planet(saturn, gas_giant, large, cold, 6th_closest_to_sun).

planet(uranus, ice_giant, large, cold, 7th_closest_to_sun).
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planet(neptune, ice_giant, large, cold, 8th_closest_to_sun).

% Predicate to find planet properties

planet_properties(Name, Type, Size, Temperature, Position) :-

    planet(Name, Type, Size, Temperature, Position).

% Example Query:

% ?- planet_properties(earth, Type, Size, Temperature, Position).

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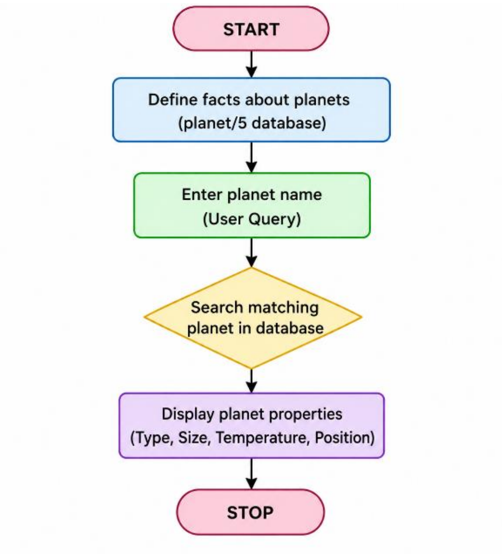
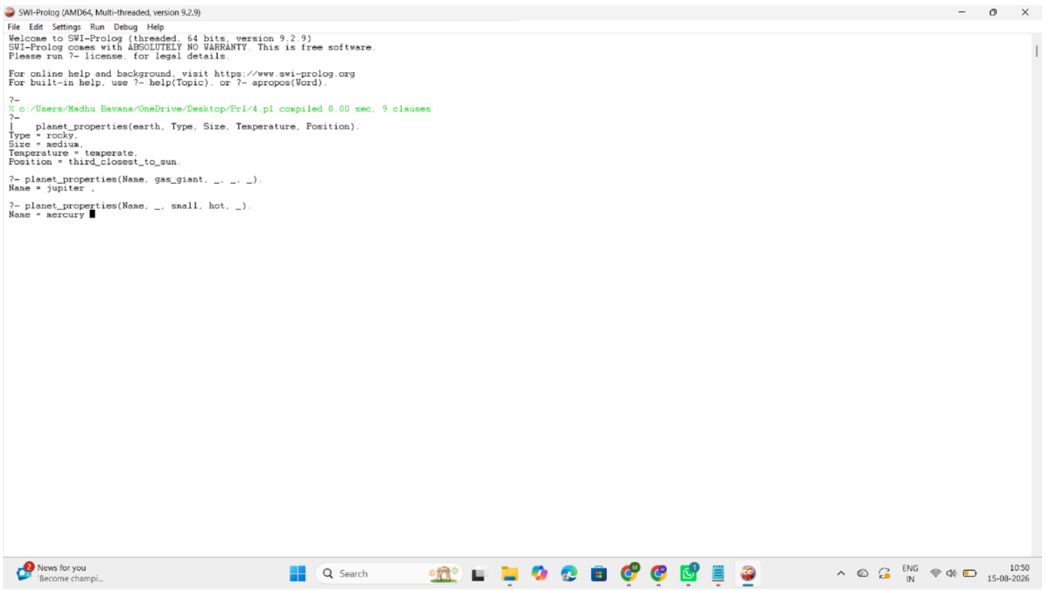


Output



Result

Thus, the **Prolog program for implementing the PLANETS Database was successfully executed**, and the required planet properties were retrieved correctly using Prolog queries.